Lift Right — Introduction

CET257 Enterprise Project — Assignment 1, Task A1 — Team Speed

1.What the Project Is

Lift Right is a browser-based, first-person 360° training simulation built with Three.js. Instead of sitting through a slideshow induction, a warehouse worker steps into a virtual warehouse, moves between the Loading Dock, Warehouse Floor and Racking Aisle, and is scored in real time on the safety decisions they make while completing a lift. The application runs as a single self-contained file in a standard desktop browser, with no server, install or plugin required, which lets a client trial it directly on a warehouse PC.

2. The Training Topic We Chose, and Why

The team selected manual handling and lifting safety in a warehouse and logistics setting as the training topic the simulation teaches. This was not the only option discussed at the first team meeting, but it was chosen for four reasons:

1. It is a genuine, high-frequency risk. Manual handling is consistently one of the leading causes of workplace injury in warehousing, and every member of the team could point to a lifting-related induction they had personally sat through.
2. It maps naturally onto a sequence. Safe lifting is a repeatable procedure — check the route, assess the load, choose a method, use correct technique, carry, place — which converts cleanly into the six-checkpoint structure the prototype needed for a scored, replayable training loop.
3. It suits a first-person 3D environment. Route-checking and stance are spatial judgements that are hard to test with a multiple-choice quiz alone but are natural to test by having the trainee actually look around, move and choose an action inside a warehouse scene.
4. It gave the team a realistic client to approach. Warehousing and logistics employers already run mandatory manual-handling inductions, which meant the team could pitch Lift Right to a genuine prospective client as a replacement for an existing training requirement rather than an entirely new one.

3. Why It Is Needed

The team's view, discussed at the first meeting and reconfirmed after the first client meeting, is that the standard approach to manual-handling induction has three recurring weaknesses. The table below sets out each problem against how Lift Right responds to it.

Problem with the current approach	**How Lift Right responds**
A one-off classroom induction or a static poster is quickly forgotten once a worker is back on the floor under time pressure.	The simulation can be repeated in five to ten minutes, so refresher training is cheap enough to actually happen rather than being a once-a-year event.
Paper sign-off sheets show that a worker attended a session, not that they can apply the technique.	Each checkpoint requires an active, scored decision — route, load assessment, handling method, technique — so a pass reflects a demonstrated judgement, not attendance.
Supervisors have no easy way to see who is	The Admin Command Center gives

actually competent, only who signed a form, especially ahead of an audit or after an incident.	supervisors per-worker login activity, attempts, time-on-task and best score, exportable to CSV for a compliance record.

4. Who Will Use It

4.1 Primary Users

- Warehouse workers ("trainees") — completing the simulation as part of new-starter induction or periodic refresher training. This group is the main audience the scene, controls and checkpoint content are designed around.

4.2 Secondary Users

- Supervisors / training administrators — do not play the simulation themselves in normal use, but log in to the Admin Command Center to review who has trained, to what standard, and to export records ahead of an audit.

4.3 Tertiary Users

- Health & safety / compliance staff — an indirect audience who receive the exported CSV record rather than using the application directly, and who were considered when deciding what the Admin Command Center needed to capture.

4.4 Accessibility Considerations

- Warehouse operatives vary in age, prior gaming experience and digital literacy, so the simulation was scoped to mouse-and-keyboard controls (look, W-A-S-D movement) rather than assuming a games controller or VR headset.
- The floor may include non-native English speakers, so checkpoint and quiz text is kept short and plain rather than written as dense policy language.
- The HUD (timer, score, zone) is designed to remain legible at both 1920x1080 and 1366x768, since the team could not assume every warehouse PC has a large or modern monitor.
- The team recorded remaining accessibility gaps — such as colour-blind-safe checkpoint markers and a keyboard-only alternative to mouse-look — as future work rather than claiming them as delivered in this prototype; see the Assumptions and Risks sections of the System Requirement Analysis document.